# 0/1 Knapscak problem using Backtracking

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# 0/1 Knapscak problem

Given two arrays $v[]$ and $w[]$ that represent values and weights associated with $n$ items respectively. Find out the maximum value subset(Maximum Profit) of $v[]$ such that the sum of the weights of this subset is smaller than or equal to Knapsack capacity $W$.

The backtracking based solution works better than brute force by ignoring infeasible solutions

# Search Strategy

- The search space is explored in a **depth-first manner**.
- We begin at the **start node**.
- This start node is both:
  - A **live node**
  - The **E-node** (expansion node)

# Expansion Rule

- From the current E-node, we attempt to move to a new node.
- If the move is feasible:
  - The new node becomes a **live node**.
  - It also becomes the new **E-node**.
  - The old E-node remains live.

# Infeasible Move

- If we cannot move to a new node:
  - The current E-node **dies** (no longer live).
  - We **backtrack** to the most recent live node.
  - That live node becomes the new E-node.

# Termination Condition

- The search terminates when:
  - A feasible solution (answer) is found, or
  - There are no live nodes left to backtrack to.
- At this point, the best solution encountered is the **optimal solution**.

# Knapsack problem-1

- Number of items: $n = 3$
- Weights: $w = [20, 15, 15]$
- Profits: $p = [40, 25, 25]$
- Capacity: $c = 30$

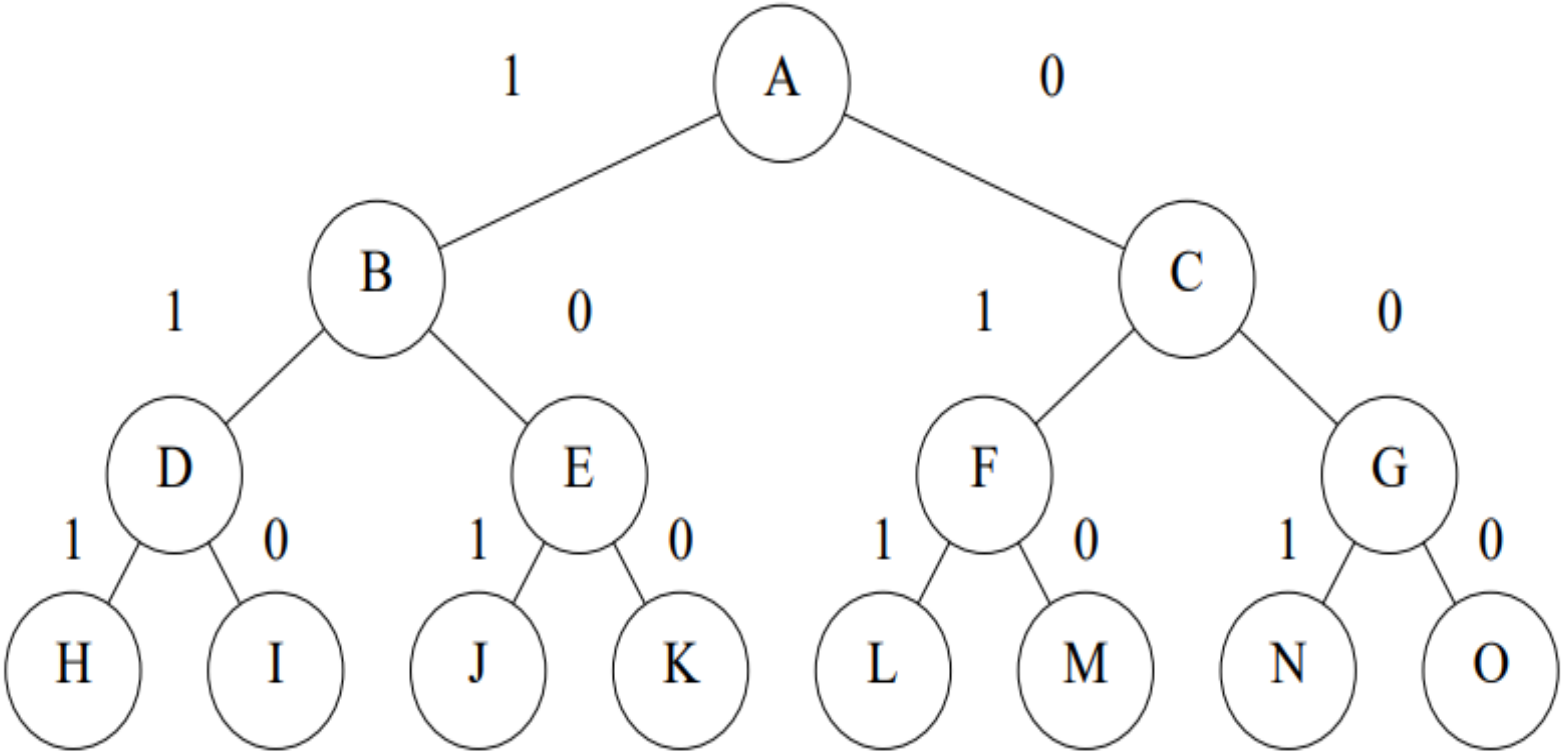


**Figure 21.2** Solution space for a three-object knapsack

# Root Node (A)

- Start at root node A.
- A is both a **live node** and the **E-node**.
- Choices:
  - Move left to B (include item 1).
  - Move right to C (exclude item 1).

# Node B: Include Item 1

- At B: capacity used = 20, profit = 40.
- Choices:
  - D (include item 2).
  - E (exclude item 2).

# Node D: Items 1 and 2

- Capacity used = 35(20item1 profit +15 item 2 profit), exceeds limit (30).
- Infeasible solution.
- Node D dies, backtrack to B.

# Node E: Item 1 Only

- Remaining capacity = 10(30-20item1profit), profit = 40.
- Choices:
  - J (include item 3).
  - K (exclude item 3).

# Node J vs Node K

- J: Items 1 and 3 $\rightarrow$ capacity 35, infeasible.
- K: Item 1 only $\rightarrow$ capacity 20, profit 40.
- Feasible leaf solution: profit $= 40$.
- Current best solution.

# Backtracking

- Node K dies, backtrack to E.
- E dies, backtrack to B.
- B dies, A becomes E-node again.
- Explore right branch: node C.

# Node C: Exclude Item 1

- Capacity = 30, profit = 0.
- Choices:
  - F (include item 2).
  - G (exclude item 2).

# Node F: Include Item 2

- Capacity =(30-15item2)=15, profit =(0+25item2)= 25.
- Choices:
  - L (include item 3).
  - M (exclude item 3).

# Node L: Items 2 and 3

- Capacity used =(15+15item3)=30, profit =(25+25item3)=50.
- Feasible leaf solution.
- Better than previous best (40).
- Update best solution: profit = 50.

# Node M and Node G

- M: Item 2 only $\rightarrow$ profit $= 25$ (less than 50).
- G: Exclude item 2 $\rightarrow$ profit $= 0$.
- From G:
  - N (include item 3) $\rightarrow$ profit $= 25$.
  - O (exclude item 3) $\rightarrow$ profit $= 0$.
- None better than 50.

# Final Result

- All nodes explored.
- Best solution: profit $= 50$.
- Items selected: 2 and 3.
- Path: $A \rightarrow C \rightarrow F \rightarrow L$.
- Knapsack filled exactly to capacity.

# Problem2

- Weights: $[2, 3, 4, 5, 9]$
- Profits: $[3, 4, 5, 8, 10]$
- Capacity: 10

**Step-by-step Backtracking:**

1. Start with item 1 (w=2, p=3). Choices: include (1) or exclude (0).
2. If included: remaining capacity = 8. Explore item 2.
   - Include item 2 (w=3, p=4): remaining capacity = 5.
     - Include item 3 (w=4, p=5): total weight = 9, profit = 12. Cannot add more.
     - Exclude item 3, include item 4 (w=5, p=8): total weight = 10, profit = 15. **Best so far**.
   - Exclude item 2: try item 3 (w=4, p=5). Profit = 8 (less).
3. If item 1 excluded: capacity = 10.
   - Include item 2 (w=3, p=4), then item 3 (w=4, p=5): total weight = 7, profit = 9.

**Optimal Solution:** Items $\{1, 2, 4\}$ with total weight 10 and profit **15**.

# Problem Setup

- Number of items: $n = 5$
- Weights: $[2, 3, 4, 5, 9]$
- Profits: $[3, 4, 5, 8, 10]$
- Capacity: $c = 10$

# Step 1: Root Node

- Start at root (capacity $= 10$, profit $= 0$).
- Decision: include item 1 (left branch) or exclude item 0 (right branch).
- Root is both a **live node** and the **E-node**.
- If item 1 included $\rightarrow$ capacity $= 8$, profit $= 3$.
- If item 1 excluded $\rightarrow$ capacity $= 10$, profit $= 0$.

# Step 2: Include Item 1

- Include item 2 → capacity =8-3item2=5, profit =3+4item2=7.
  - Include item 3 → capacity = 1, profit = 12 (cannot add item 4).
  - Exclude item 3, include item 4 → capacity =5-5item4=0, profit = 15 (**feasible leaf**).
- Exclude item 2 → capacity = 8, profit = 3.
- Explore items (3, 4, 5). All combinations are considered:

| Items chosen | Total weight | Total profit |
|---|---|---|
| 3 only | $2 + 4 = 6$ | $3 + 5 = 8$ |
| 4 only | $2 + 5 = 7$ | $3 + 8 = 11$ |
| 5 only | $2 + 9 = 11$ (infeasible) | – |
| 3 and 4 | $2 + 4 + 5 = 11$ (infeasible) | – |
| 3 and 5 | $2 + 4 + 9 = 15$ (infeasible) | – |
| 4 and 5 | $2 + 5 + 9 = 16$ (infeasible) | – |

Table: Exploration of items 3, 4, 5 after excluding item 2

- Best feasible profit in this branch = 11 (items 1 and 4).
- Compare with global best (15 from items 1, 2, 4).

# Step 3: Exclude Item 1

- Capacity = 10, profit = 0.
- Explore items (2, 3, 4, 5). All combinations are considered:

| Items chosen | Total weight | Total profit | Feasible? |
|---|---|---|---|
| 2 only | 3 | 4 | Yes |
| 3 only | 4 | 5 | Yes |
| 4 only | 5 | 8 | Yes |
| 5 only | 9 | 10 | Yes |
| 2 and 3 | $3 + 4 = 7$ | $4 + 5 = 9$ | Yes |
| 2 and 4 | $3 + 5 = 8$ | $4 + 8 = 12$ | Yes |
| 2 and 5 | $3 + 9 = 12$ | $4 + 10 = 14$ | No |
| 3 and 4 | $4 + 5 = 9$ | $5 + 8 = 13$ | Yes |
| 3 and 5 | $4 + 9 = 13$ | $5 + 10 = 15$ | No |
| 4 and 5 | $5 + 9 = 14$ | $8 + 10 = 18$ | No |

Table: Exploration of items 2, 3, 4, 5 when Item 1 is excluded

- Best feasible profit in this branch = **13** (items 3 and 4).
- Global best remains **15** (items 1, 2, 4).

# Optimal Solution

- Path: Include items 1, 2, and 4.
- Capacity used = 10.
- Profit = **15**.
- This is the **optimal solution**.

# Problem 2: 0/1 Knapsack Backtracking

**Given:**

- Weights: $[2, 3, 4, 5, 9]$
- Profits: $[3, 4, 5, 8, 10]$
- Capacity: 10

# All combinations-Step-by-step Backtracking Exploration

| Items chosen | Total weight | Total profit | Feasible? |
|---|---|---|---|
| 1,2,3 | $2+3+4=9$ | $3+4+5=12$ | Yes |
| 1,2,4 | $2+3+5=10$ | $3+4+8=15$ | **Yes (Optimal)** |
| 1,2,5 | $2+3+9=14$ | $3+4+10=17$ | No |
| 1,3,4 | $2+4+5=11$ | $3+5+8=16$ | No |
| 1,3 | $2+4=6$ | $3+5=8$ | Yes |
| 1,4 | $2+5=7$ | $3+8=11$ | Yes |
| 2,3 | $3+4=7$ | $4+5=9$ | Yes |
| 2,4 | $3+5=8$ | $4+8=12$ | Yes |
| 3,4 | $4+5=9$ | $5+8=13$ | Yes |
| 5 only | 9 | 10 | Yes |

Table: Exploration of feasible combinations using backtracking

- Best feasible profit = **15** with items $\{1, 2, 4\}$.
- This is the optimal solution.

# 0/1 Knapsack Algorithm (Backtracking)

**Algorithm:**

1. Start at the root node with capacity $M$ and profit 0.
2. At each level $i$, decide:
   - Include item $i$ (go left).
   - Exclude item $i$ (go right).
3. Update capacity and profit accordingly.
4. If capacity $< 0 \rightarrow$ prune branch (infeasible).
5. If capacity $= 0$ or all items considered $\rightarrow$ record solution.
6. Keep track of maximum profit found.

# Worked Example

**Problem:**

- Weights: $[2, 3, 4, 5, 9]$
- Profits: $[3, 4, 5, 8, 10]$
- Capacity: 10

**Step-by-step:**

1. Root: capacity 10, profit 0.
2. Include item 1 (w=2, p=3) $\rightarrow$ capacity 8, profit 3.
3. Include item 2 (w=3, p=4) $\rightarrow$ capacity 5, profit 7.
4. Include item 3 (w=4, p=5) $\rightarrow$ capacity 1, profit 12 (cannot add item 4).
5. Backtrack: exclude item 3, include item 4 (w=5, p=8) $\rightarrow$ capacity 0, profit 15.
6. Record solution: items $\{1, 2, 4\}$, profit 15.
7. Explore other branches (e.g., exclude item 1 $\rightarrow$ best profit 13).